Calibration results
===================
Normalized Residuals

Reprojection error (cam0):   mean 2.1459664999152115, median 1.5135842609854728, std: 2.029047909892085
Reprojection error (cam1):   mean 2.2552968456496356, median 1.680939513380745, std: 2.146248276327054
Gyroscope error (imu0):   mean 1.1420015122942646, median 0.7952115558780923, std: 1.5995312327376252
Accelerometer error (imu0):   mean 2.669588421673703, median 1.9407645932690492, std: 3.6714133990019238

Residuals

Reprojection error (cam0) [px]:   mean 2.1459664999152115, median 1.5135842609854728, std: 2.029047909892085
Reprojection error (cam1) [px]:   mean 2.2552968456496356, median 1.680939513380745, std: 2.146248276327054
Gyroscope error (imu0) [rad/s]:   mean 0.0006551005847488037, median 0.0004561671325707583, std: 0.000917559070289907
Accelerometer error (imu0) [m/s^2]: mean 0.01891440756428274, median 0.01375058874446489, std: 0.026012477729489784

Transformation (cam0):

T_ci: (imu0 to cam0):
[[0.13633614 -0.99055455 0.01463361 0.02990328]
[0.034182   -0.01005909 -0.999365   0.01511571]
[0.99007275 0.13674978 0.03248771 -0.10593778]
[0.   0.   0.   1.   ]]

T_ic: (cam0 to imu0):
[[0.13633614 0.034182   0.99007275 0.10029253]
[-0.99055455 -0.01005909 0.13674978 0.04425985]
[0.01463361 -0.999365   0.03248771 0.01811019]
[0.   0.   0.   1.   ]]

timeshift cam0 to imu0: [s] (t_imu = t_cam + shift)
-0.14230599579312334

Transformation (cam1):

```
-----------------------
T_ci:  (imu0 to cam1):
[[-0.04221275 -0.99872481  0.02769177 -0.03723357]
 [ 0.03583804 -0.02921222 -0.99893057  0.01285847]
 [ 0.99846568 -0.04117519  0.03702547 -0.1073803 ]
 [ 0.          0.          0.          1.        ]]

T_ic:  (cam1 to imu0):
[[-0.04221275  0.03583804  0.99846568  0.10518299]
 [-0.99872481 -0.02921222 -0.04117519 -0.04123187]
 [ 0.02769177 -0.99893057  0.03702547  0.01785159]
 [ 0.          0.          0.          1.        ]]

timeshift cam1 to imu0: [s] (t_imu = t_cam + shift)
-0.1934082413626727

Baselines:
----------
Baseline (cam0 to cam1):
[[ 0.98394151 -0.01907083 -0.17746945 -0.0851691 ]
 [ 0.01920436  0.99981511 -0.00096547 -0.00293099]
 [ 0.17745505 -0.00245822  0.98412584 -0.00839352]
 [ 0.          0.          0.          1.        ]]
baseline norm:  0.0856318717360083 [m]


Gravity vector in target coords: [m/s^2]
[-4.17205432 -0.06451686 -8.87458299]


Calibration configuration
=========================

cam0
-----
```

Camera model: pinhole
Focal length: [157.90718294969474, 150.82613798877807]
Principal point: [137.2392017892296, 162.09926651723032]
Distortion model: radtan
Distortion coefficients: [-0.03144610780506922, -0.009209428514045158, 0.00809137604018658, -0.024315044913275994]
Type: checkerboard
Rows
Count: 8
Distance: 0.02 [m]
Cols
Count: 11
Distance: 0.02 [m]

cam1

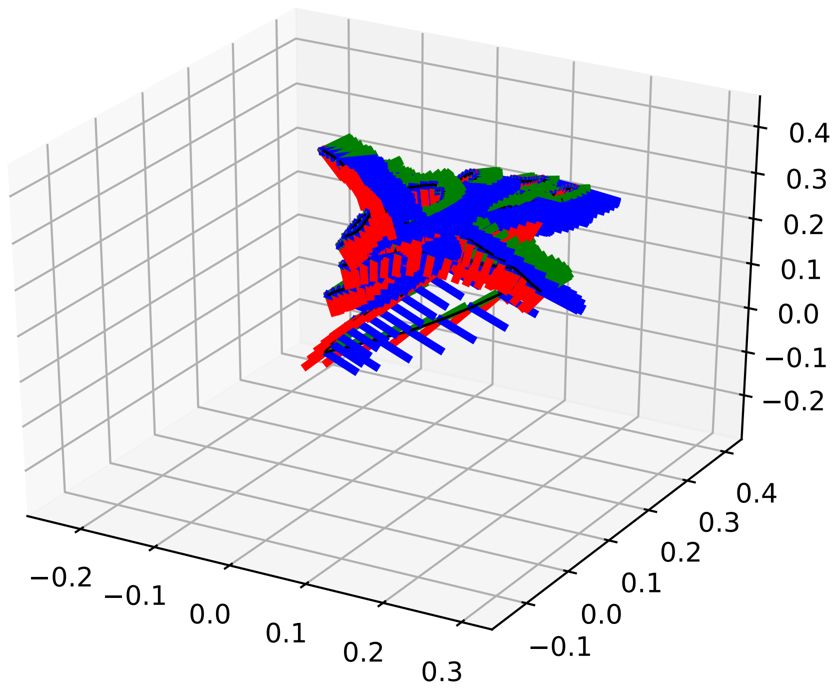
Camera model: pinhole
Focal length: [153.68049730029296, 147.35324765779288]
Principal point: [174.0901722806695, 160.73563396517]
Distortion model: radtan
Distortion coefficients: [-0.07339047497350926, -0.0015845181685253686, 0.017515404728075147, 0.0032496851992207887]
Type: checkerboard
Rows
Count: 8
Distance: 0.02 [m]
Cols
Count: 11
Distance: 0.02 [m]

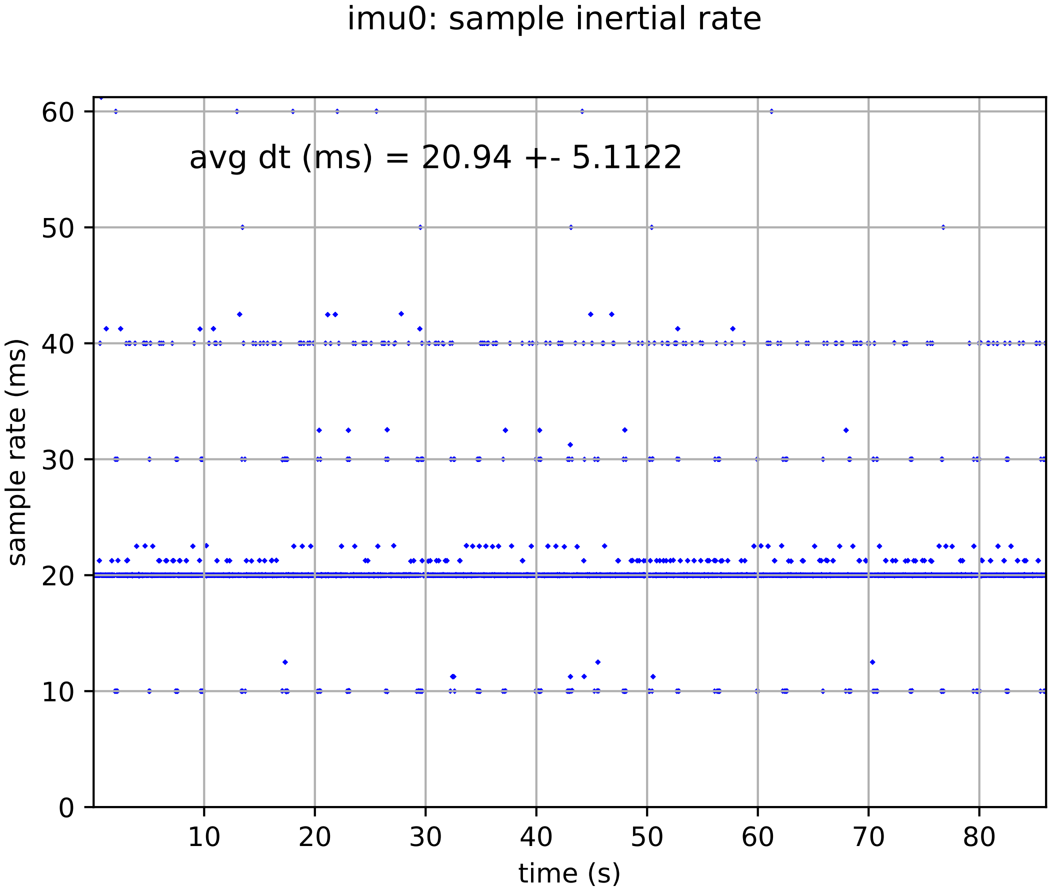
IMU configuration
=================

IMU0:

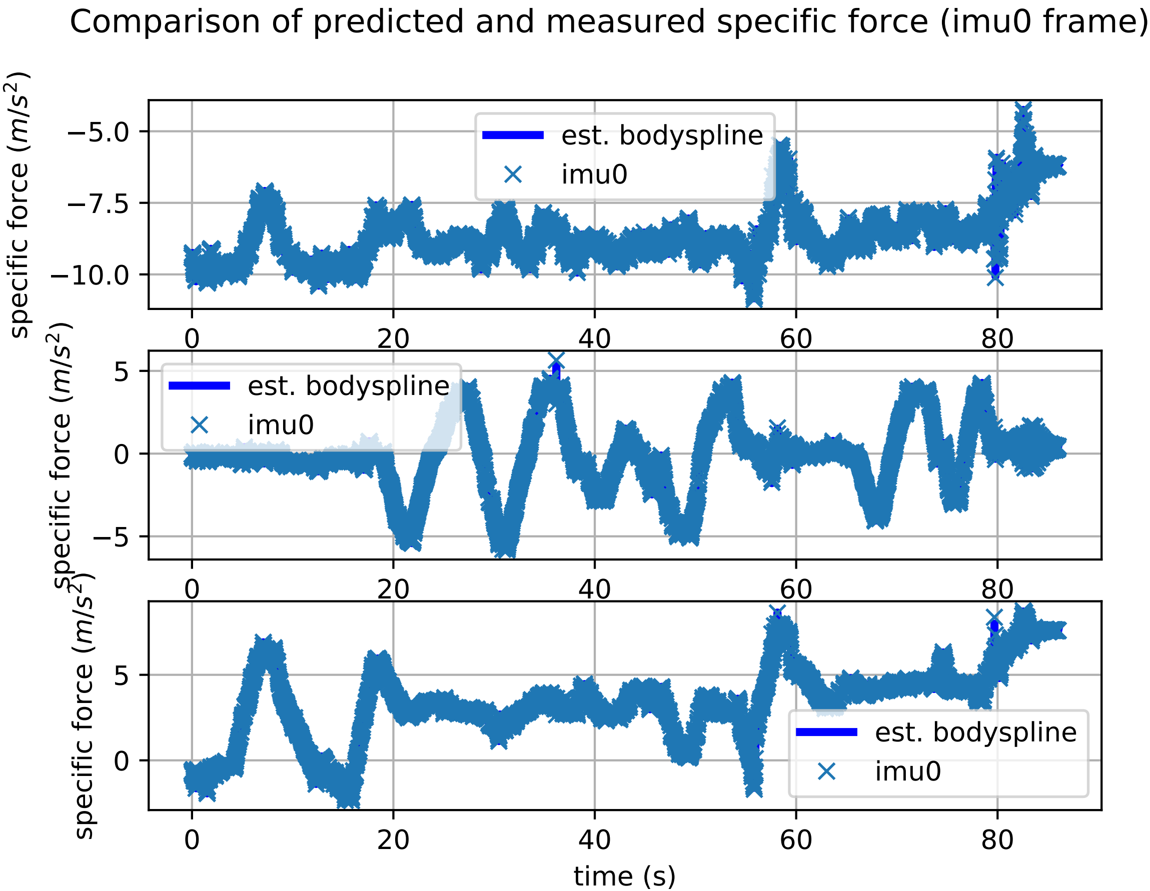
Model: calibrated
Update rate: 50.0
Accelerometer:
Noise density: 0.0010019901002151698
Noise density (discrete): 0.007085139945439349
Random walk: 1.9893785362909353e-05
Gyroscope:
Noise density: 8.112529814510266e-05
Noise density (discrete): 0.0005736424844418254
Random walk: 6.28721982800146e-07
T_ib (imu0 to imu0)
[[1. 0. 0. 0.]
[0. 1. 0. 0.]
[0. 0. 1. 0.]
[0. 0. 0. 1.]]
time offset with respect to IMU0: 0.0 [s]

imu0: estimated poses

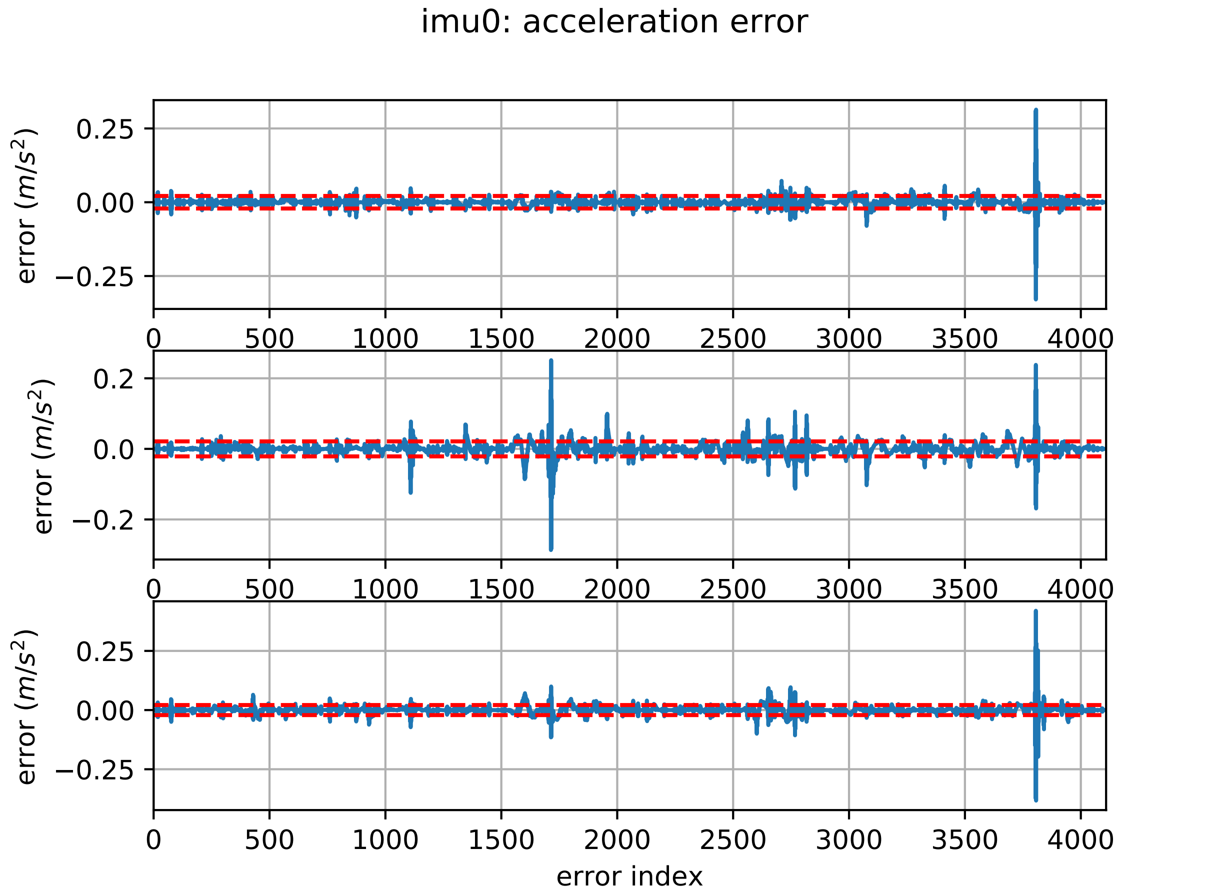


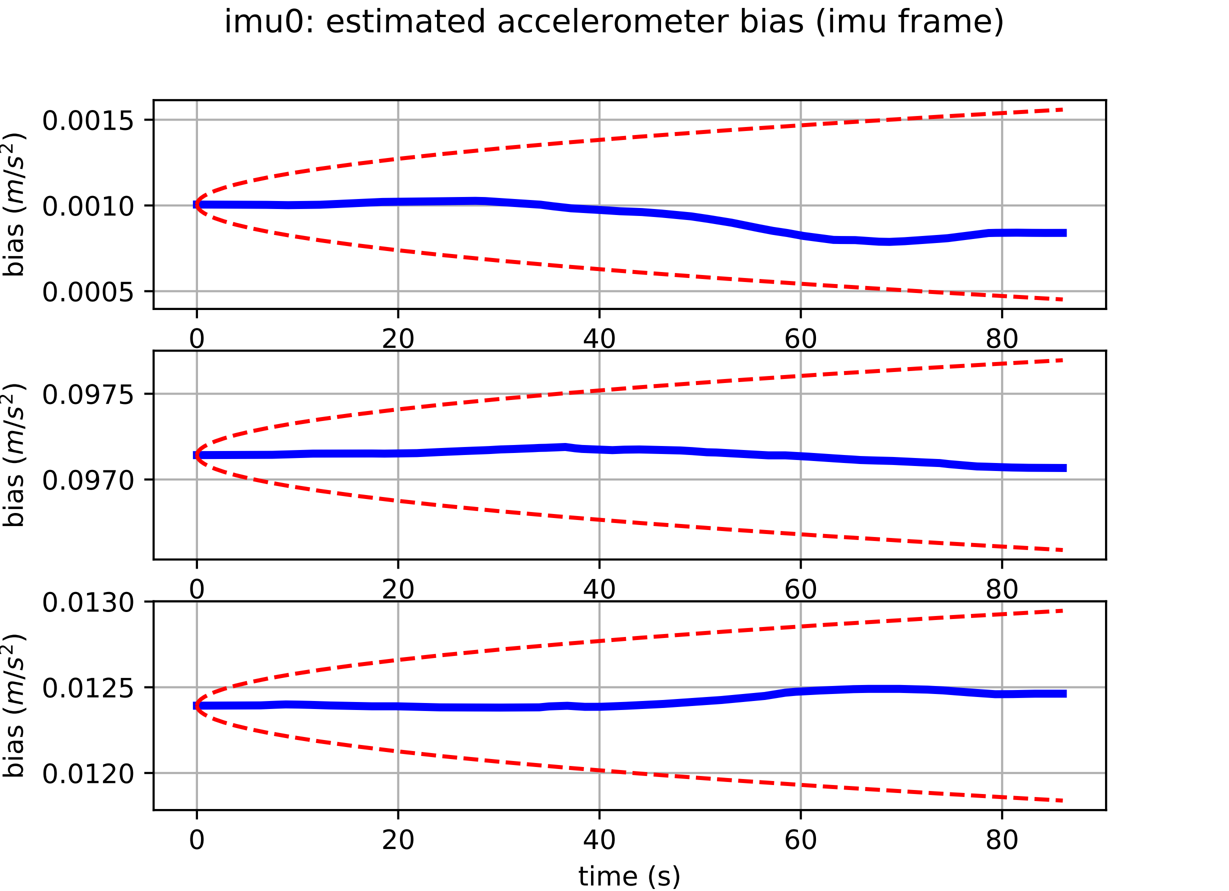


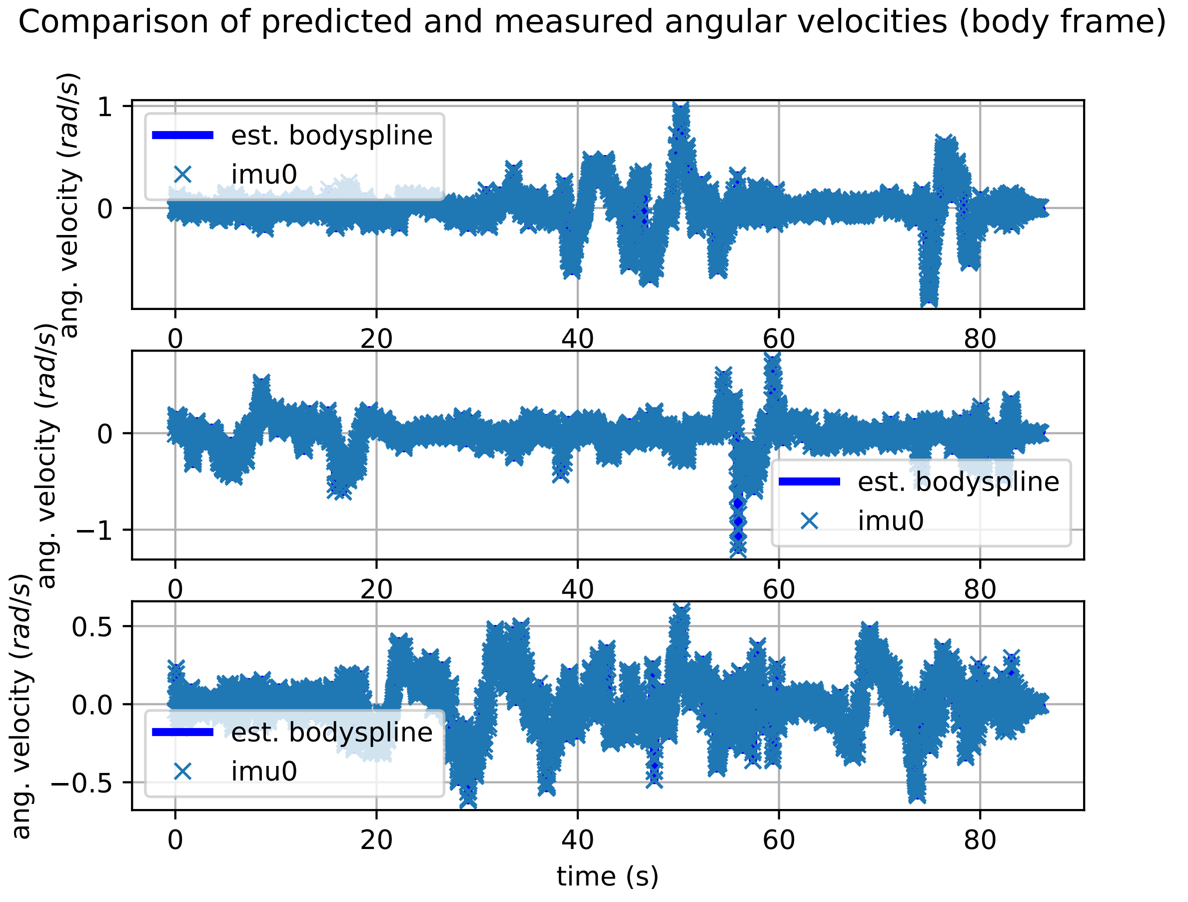
imu0: sample inertial rate
avg dt (ms) = 20.94 +- 5.1122
sample rate (ms)
time (s)
0
10
20
30
40
50
60
10
20
30
40
50
60
70
80

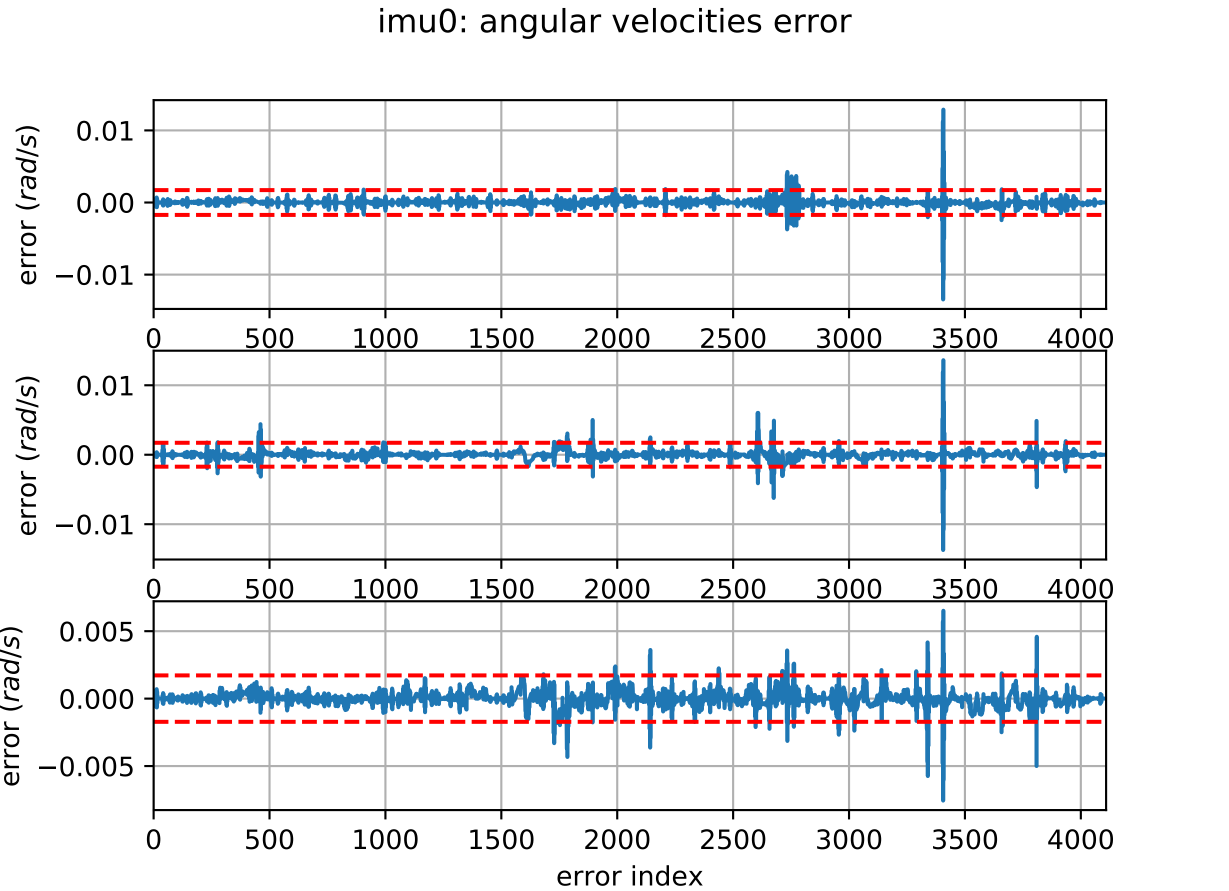


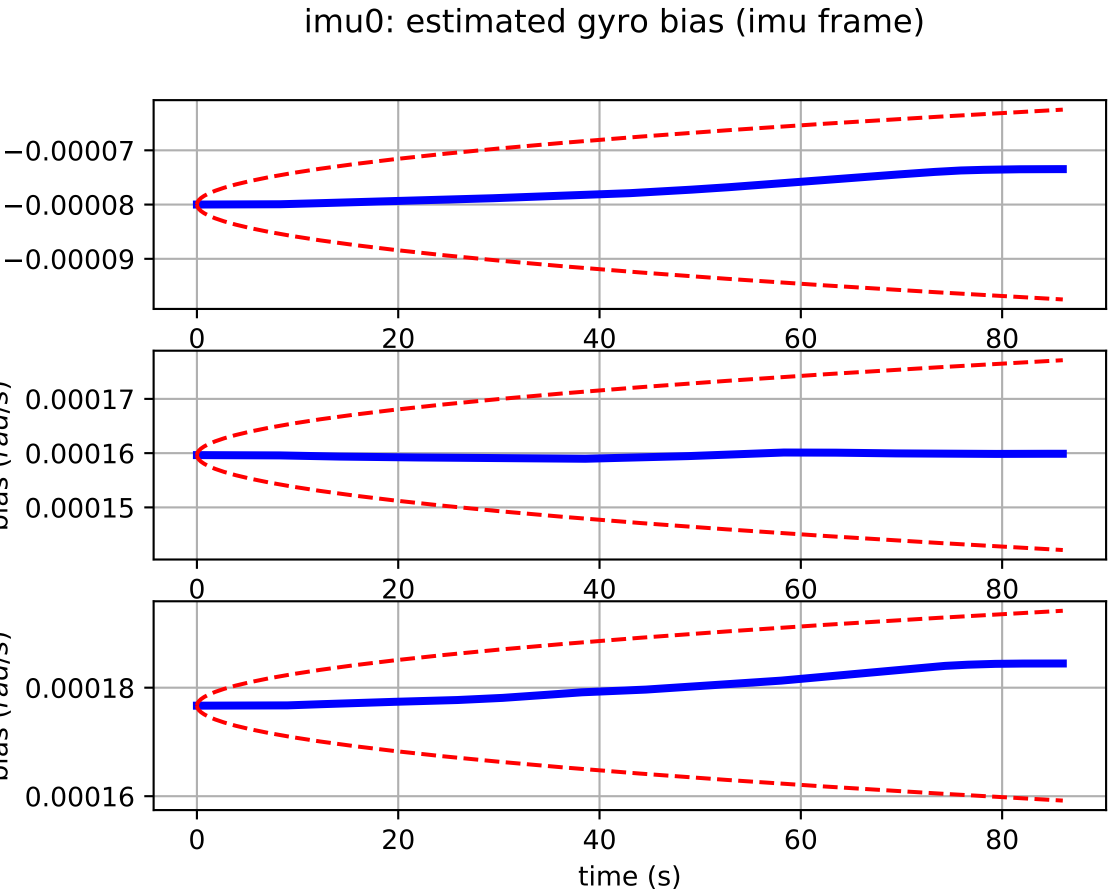
Comparison of predicted and measured specific force (imu0 frame)
specific force (m/s^2)
−5.0
−7.5
−10.0
est. bodyspline
imu0
0
20
40
60
80
specific force (m/s^2)
5
0
−5
est. bodyspline
imu0
0
20
40
60
80
specific force (m/s^2)
5
0
est. bodyspline
imu0
0
20
40
60
80
time (s)

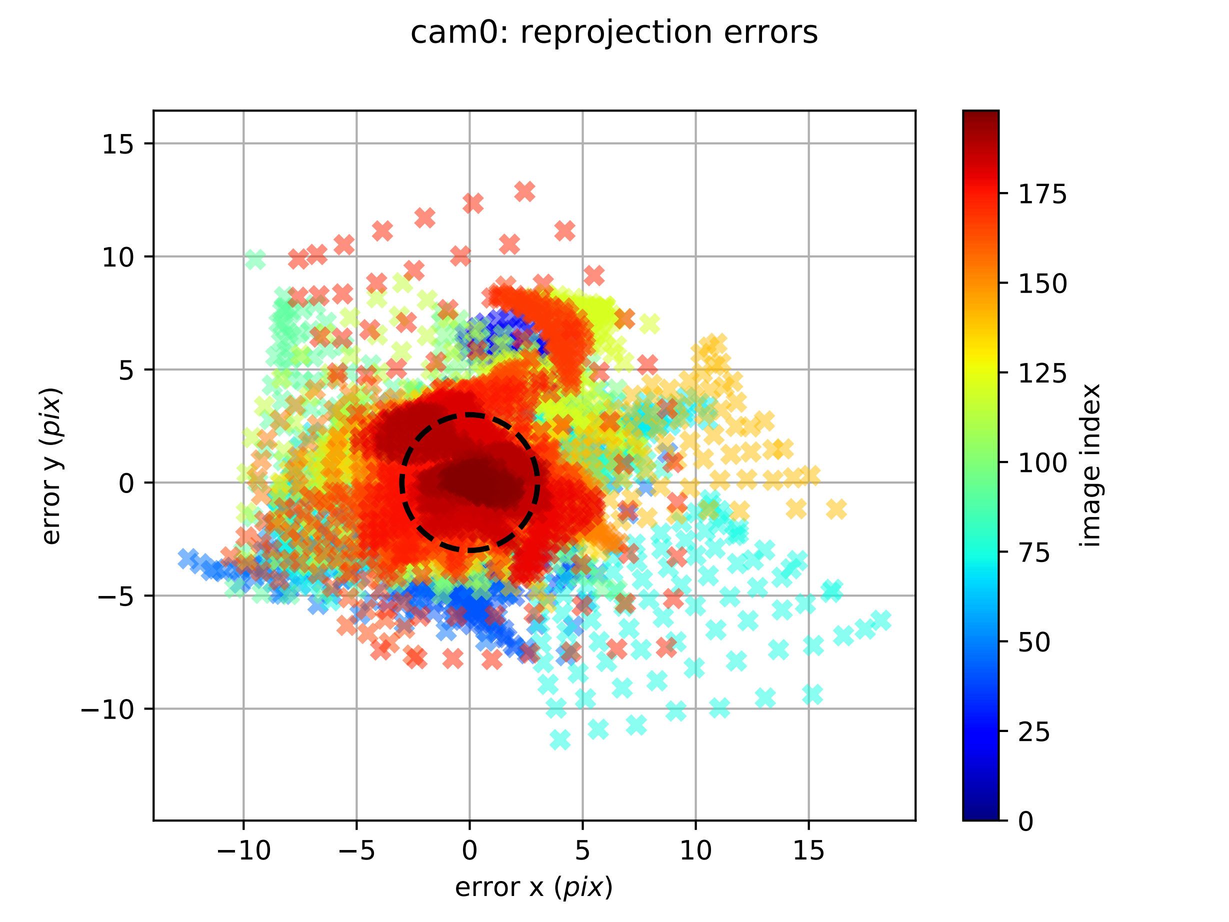

imu0: acceleration error
error (m/s^2)
0.25
0.00
−0.25
0.2
0.0
−0.2
0
500
1000
1500
2000
2500
3000
3500
4000
error index

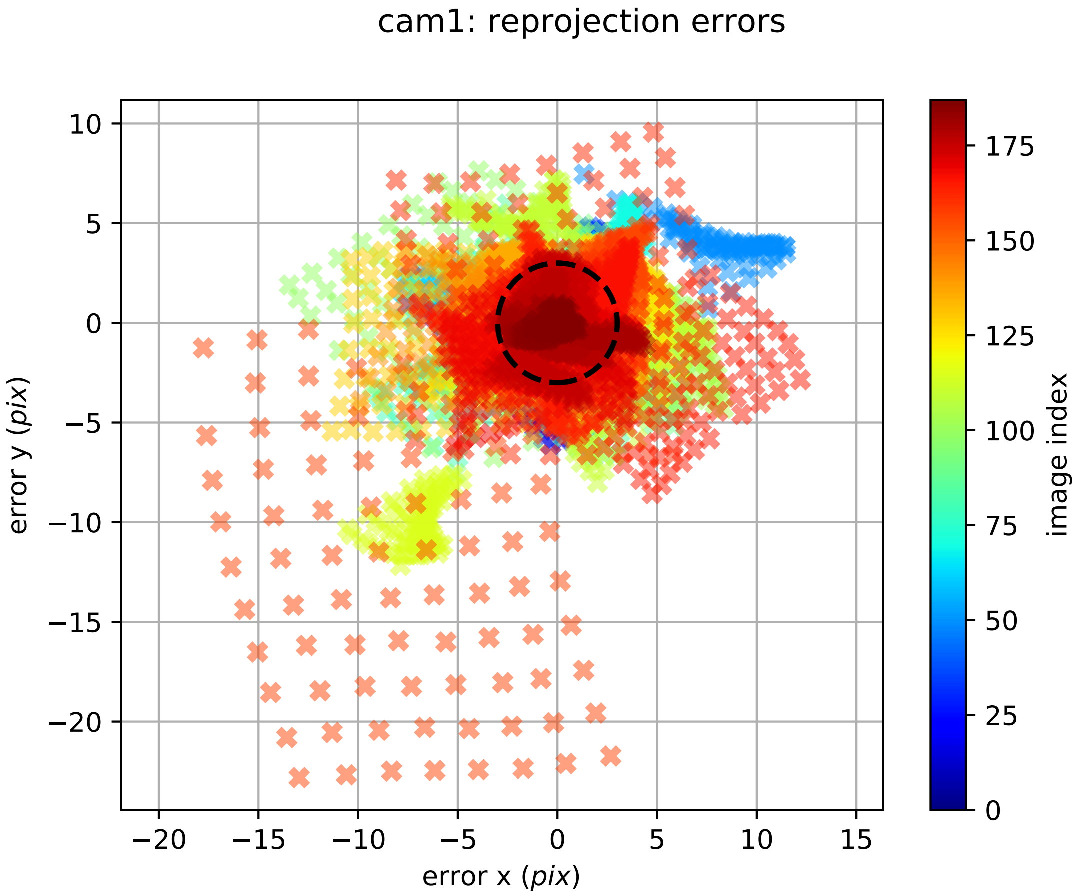

imu0: estimated accelerometer bias (imu frame)
bias (m/s^2)
0.0015
0.0010
0.0005
0.0975
0.0970
0.0130
0.0125
0.0120
0
20
40
60
80
time (s)


Comparison of predicted and measured angular velocities (body frame)
ang. velocity (rad/s)
est. bodyspline
imu0
1
0
0
20
40
60
80
ang. velocity (rad/s)
0
−1
est. bodyspline
imu0
0
20
40
60
80
ang. velocity (rad/s)
0.5
0.0
−0.5
est. bodyspline
imu0
0
20
40
60
80
time (s)


imu0: angular velocities error
error (rad/s)
0.01
0.00
−0.01
0.005
0.000
−0.005
0
500
1000
1500
2000
2500
3000
3500
4000
error index


imu0: estimated gyro bias (imu frame)
−0.00007
−0.00008
−0.00009
0.00017
0.00016
0.00015
0.00018
0.00016
0
20
40
60
80
time (s)


cam0: reprojection errors
error x (pix)
error y (pix)
image index
−10
−5
0
5
10
15
175
150
125
100
75
50
25
0


cam1: reprojection errors
error y (pix)
error x (pix)
image index
10
5
0
−5
−10
−15
−20
−20
−15
−10
−5
0
5
10
15
175
150
125
100
75
50
25
0